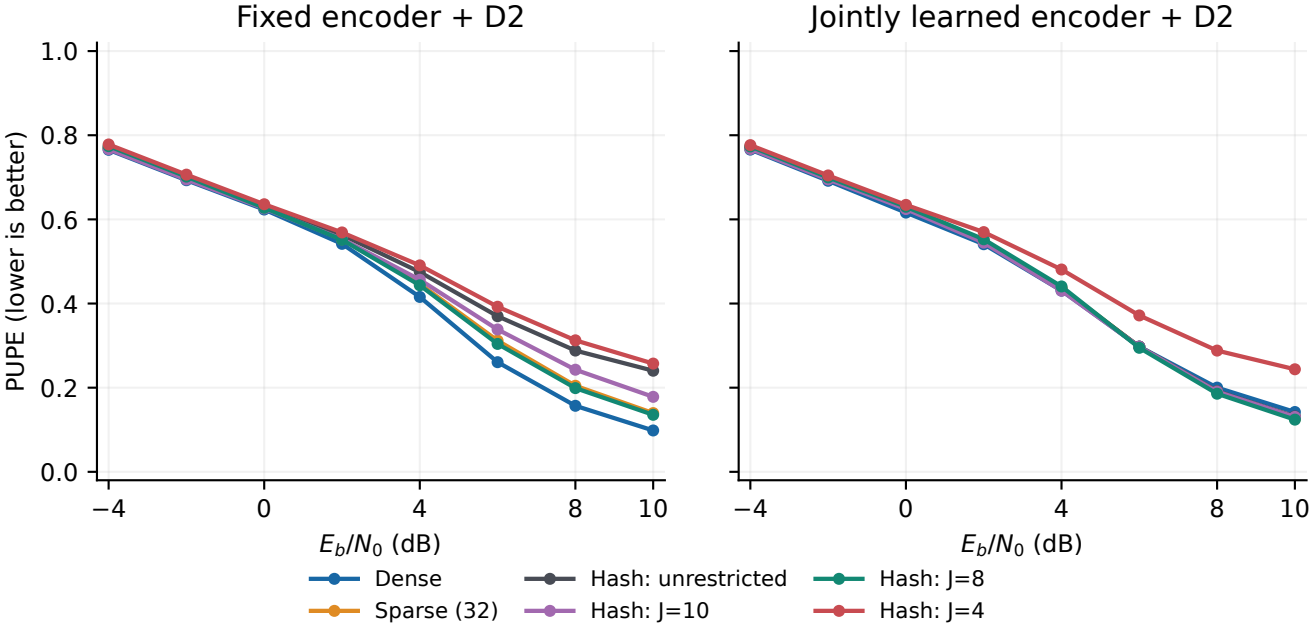

Fixed encoder + D2
Jointly learned encoder + D2
PUPE (lower is better)
1.0
0.8
0.6
0.4
0.2
0.0
−4
0
4
8
10
E_b/N_0 (dB)
Dense
Sparse (32)
Hash: unrestricted
Hash: J=10
Hash: J=8
Hash: J=4